

FDDT Case Report

Fraud Document Detection Tool — system-assisted analysis for human review. See Section 6 for what this report is and is not.

1. Case Details

Case ID	CASE-DE627FA2 de627fa2-3d99-4340-8b3e-79ad041bb7d4
Submitted by	test1Reviewer L2
Submission date	2026-10-06 15:07 UTC
Documents	1 Types: School document
Case type	Procurement documentation
Status / decision	Pending manual review / Pending review
Report generated	2026-10-06 15:12 UTC by test1Reviewer L2 (Reviewer L2)
Report ID	377efb18-1ba6-4452-b5bb-50378917ba86

2. Executive Summary

HIGH RISK Score 90 / 100 (computed 2026-10-06 15:08 UTC)

Why: Driven mainly by (40 points): 'case2.pdf': 1 page(s) are near-identical to previously submitted documents. Page 1 is a near-identical match (perceptual hash Hamming distance 0) to page 1 of "sample21.pdf" in case CASE-CF609781 —...

Decision status: Pending review.

Exceptions: 5 item(s) did not pass, 4 of them shown as highlighted regions on the source pages. Section 5 lists each one; Section 9 shows them on the documents.

This summary is deliberately short. The reasons behind the tier are in Section 3, the complete checklist is in Section 4.

3. Explainable Findings

The risk engine is a transparent, weighted rule set. Each row is one rule that fired, with the reason text stored at scoring time. Points are summed (raw 90), capped at 100, and mapped to a tier (medium from 30, high from 60) — here **90/100, high**.

Severity	Points	Check	Document / page	Reason
HIGH	40	Duplicate detection	case2.pdf · page 1	'case2.pdf': 1 page(s) are near-identical to previously submitted documents. Page 1 is a near-identical match (perceptual hash Hamming distance 0) to page 1 of "sample21.pdf" in case CASE-CF609781 — a possible duplicate or near-duplicate resubmission. Highlights: R2
MEDIUM	15	Field validation	case2.pdf · page 1	'case2.pdf': Tax 42.50 is not 5% of the total less tax 57,140.00 (expected 2,857.00). Highlights: R3, R4
MEDIUM	15	Issuer verification	case2.pdf	Issuer 'أكاديمية الخليج الدولية الخاصة' THE GULF INTERNATIONAL PRIVATE ACADEMYO' on 'case2.pdf' did not match any known issuer in the registry.
MEDIUM	10	Metadata forensics	case2.pdf	'case2.pdf' embeds JavaScript or an open-action — unusual for a business document.
MEDIUM	10	Visual inconsistency / AI-generation review	case2.pdf · page 1	'case2.pdf': the vision-model review reports misaligned text. Page 1 — Text alignment: The signatures at the bottom are misaligned and have different baselines compared to the typed text around them, notably in the Registration Department and Accounts Department areas (approx. bounding box x:0.1, y:0.8, width:0.4, height:0.15). This is a vision-model judgment, not pixel-level analysis — a probabilistic signal, not a definitive finding. Highlights: R1

4. All Checks

The completeness record: every check that ran on every document, including those that passed. Section 3 is the narrative; this is the checklist.

Document 1 — case2.pdf (School document · 4 passed, 5 flagged)

Check	Result	Details
Metadata forensics PDF metadata, edit history and incremental-save markers.	Flag	1 finding(s). This PDF contains an OpenAction and/or embedded JavaScript — unusual for a business document.
Font consistency Text set in a different font family from the text around it (text layer, or OCR font recognition on scans).	Pass	No anomaly detected by this check.
Error level analysis (ELA) Pixel-level recompression differences that can indicate edited regions.	Pass	No anomaly detected by this check.
Copy-move detection Regions of a page that were duplicated elsewhere on the same page.	Pass	No anomaly detected by this check.
Visual inconsistency / AI-generation review Model review of fonts, alignment, contrast, sharpness, lighting and signs of AI generation.	Flag	1 finding(s). Page 1 — Text alignment: The signatures at the bottom are misaligned and have different baselines compared to the typed text around them, notably in the Registration Department and Accounts Department areas (approx. bounding box x:0.1, y:0.8, wi...
Signature / stamp detection Whether a signature or stamp mark is present where this kind of document normally carries one.	Pass	3 signature/stamp region(s) located (presence only).
Duplicate detection Perceptual-hash comparison of each page against previously submitted pages.	Flag	1 finding(s). Page 1 is a near-identical match (perceptual hash Hamming distance 0) to page 1 of "sample21.pdf" in case CASE-CF609781 — a possible duplicate or near-duplicate resubmission.
Field validation Date, subtotal / tax / total and reference-number plausibility.	Flag	Tax 42.50 is not 5% of the total less tax 57,140.00 (expected 2,857.00).
Issuer verification Issuer name matched against the issuer registry.	Flag	Issuer 'أكاديمية الخليج الدولية الخاصة' THE GULF INTERNATIONAL PRIVATE ACADEMYO' did not match any known issuer in the registry.

5. Exceptions

Everything that did **not** pass, drawn from Section 4 and written as concrete items (5 in total). A region ID (R1, R2, ...) points to the highlight that shows the exception on the source page in Section 9 — click it to jump there.

No.	Exception	Where	Highlight
1	Metadata forensics: This PDF contains an OpenAction and/or embedded JavaScript — unusual for a business document.	Document 1 (case2.pdf)	—
2	Visual review: text alignment inconsistency (model judgment) Page 1 — Text alignment: The signatures at the bottom are misaligned and have different baselines compared to the typed text around them, notably in the Registration Department and Accounts Department areas (approx. bounding box x:0.1, y:0.8, width:0.4, height:0.15). This is a vision-model judgment, not pixel-level an...	Document 1 (case2.pdf) · page 1	R1
3	Duplicate: page is a near-duplicate of an earlier submission Page 1 is a near-identical match (perceptual hash Hamming distance 0) to page 1 of "sample21.pdf" in case CASE-CF609781 — a possible duplicate or near-duplicate resubmission.	Document 1 (case2.pdf) · page 1	R2
4	Tax / rate miscalculation: Tax 42.50 is not 5% of the total less tax 57,140.00 (expected 2,857.00).	Document 1 (case2.pdf) · page 1	R3, R4
5	Issuer not in registry: Issuer 'أكاديمية الخليج الدولية الخاصة THE GULF INTERNATIONAL PRIVATE ACADEMYO' did not match any known issuer in the registry.	Document 1 (case2.pdf)	—

2 exception(s) show “—” because they have no place on a page (for example a missing field, or a check that did not localize its finding).

6. Limitations and Assumptions

Please read this section before relying on anything in this report.

Limitations of This Analysis

What this report is

This is **system-assisted analysis to support human review**. It is not a certified forensic or legal determination that any document is genuine, forged or fraudulent, and it must not be presented as one.

Three kinds of finding

Deterministic pixel and structure checks — error level analysis, copy-move detection and PDF metadata forensics — are computed by fixed algorithms over the file’s pixels and structure. Localized findings are drawn with **solid** boxes (ELA in red, copy-move in orange). They are signals, not proof: ELA in particular can false-positive on legitimately recompressed scans and can miss print-and-rescan tampering.

Deterministic rule-based field checks — field validation and cross-document consistency — compare extracted values by fixed rules and are drawn with **solid purple** boxes. They are exact about the values they compare, but only as reliable as the OCR and extraction that produced those values.

Model-judgment checks — the visual inconsistency review, the AI-generation assessment and the signature/stamp comparison — are **probabilistic assessments** made by a vision-language model, not deterministic proof. They can vary from run to run, and their boxes are only approximate; they are the **dashed** boxes captioned “approximate” throughout this report. Signature and stamp detection shows presence and placement only and is not an identity match; the AI-generation assessment is experimental, and its accuracy on scanned business documents has not been validated.

What the system can and cannot see

The system evaluates the **submitted document file itself**. It cannot verify facts external to the document — for example whether a real-world transaction actually occurred, whether the named issuer really produced the document, or (without a pre-existing reference signature) who a signatory is. Printed Arabic and English text are read; handwritten Arabic is not reliably read.

Absence of a finding

The absence of a flagged finding does not guarantee that a document is genuine. It means only that no supported check detected an anomaly.

Check coverage on this case

Not every check applies to every document or case. This is what actually ran here:

Check	Status	Coverage on this case
Metadata forensics	Ran	Ran on 1 of 1 document(s); 1 flagged.
Font consistency	Ran	Ran on 1 of 1 document(s); 0 flagged.
Error level analysis (ELA)	Ran	Ran on 1 of 1 document(s); 0 flagged.
Copy-move detection	Ran	Ran on 1 of 1 document(s); 0 flagged.
Visual inconsistency / AI-generation review	Ran	Ran on 1 of 1 document(s); 1 flagged.
Signature / stamp detection	Ran	Ran on 1 of 1 document(s); 0 flagged.
Duplicate detection	Ran	Ran on 1 of 1 document(s); 1 flagged.
Field validation	Ran	Ran on 1 of 1 document(s); 1 flagged.
Issuer verification	Ran	Ran on 1 of 1 document(s); 1 flagged.
Signature comparison (advisory)	Did not run	Did not run — no reference signature was set for this case.
Cross-document consistency	Not applicable	Not applicable — this case has a single document.

Risk score and rule versions

The risk score in this report (**90/100, high**) was computed at 2026-10-06 15:08 UTC (assessment 8addbf56-bd0e-40b4-b551-a558d1252d92) from the risk-rule versions that fired: duplicate.cross_case_match v1 (weight 40); field.tax_rate_mismatch v1 (weight 15); issuer.not_in_registry v1 (weight 15);

metadata.javascript_or_openaction v1 (weight 10); visual.alignment_inconsistency v1 (weight 10). Tier thresholds at that time: medium from 30, high from 60.

Rule weights are versioned and are never re-applied retroactively: if an administrator later edits a rule, this case is not re-scored, and **this report remains an accurate historical record** of what was known and weighted when it was generated. A report generated later may legitimately differ.

Assumptions

What the analysis took to be true without independently verifying it. These are specific to what ran on this case.

- The issuer registry (0 active entries) was assumed to be current and accurate as of 2026-10-06 15:12 UTC. An issuer that is absent from it is reported as “not in registry”, which is a prompt to verify the issuer — not a finding that the issuer is fraudulent.
- Field values were read by OCR followed by language-model extraction and are assumed to have been read correctly. Each field carries a confidence and an “uncertain” flag. This build records that flag but does not automatically route low-confidence documents to manual review, so it is listed here for the reviewer instead: no field was marked uncertain on this case.
- Purple field highlights were located by matching extracted values back onto OCR word positions; treat their position as approximate. The comparison behind each one uses the stored, normalized values, not the highlight.
- The SHA-256 chain of custody shows the stored file is unchanged since upload. It does not show that the uploaded file is the issuer’s original — a document altered before upload is judged on its content, like any other.
- Risk-rule weights, severities and tier thresholds are administrator-configured and assumed to be appropriately calibrated; the score is advisory and is not a probability of fraud.

7. Audit Trail

Chronological case-level events, read from the append-only audit log (times in UTC). This report does not write to or duplicate the log.

Time	Event	Actor	Detail
2026-10-06 15:07 UTC	Case created	test1Reviewer L2 (Reviewer L2)	
2026-10-06 15:07 UTC	Document uploaded	test1Reviewer L2 (Reviewer L2)	case2.pdf; SHA-256 e9c3ea682c7407ba...
2026-10-06 15:07 UTC	Metadata forensics completed	System	case2.pdf
2026-10-06 15:07 UTC	ELA and copy-move checks completed	System	case2.pdf
2026-10-06 15:07 UTC	Duplicate detection completed	System	case2.pdf
2026-10-06 15:07 UTC	Signature / stamp detection completed	System	case2.pdf
2026-10-06 15:07 UTC	Font consistency check completed	System	case2.pdf
2026-10-06 15:07 UTC	Text extraction, classification and field extraction completed	System	case2.pdf
2026-10-06 15:07 UTC	Field validation and issuer verification completed	System	case2.pdf
2026-10-06 15:08 UTC	Visual inconsistency review completed	System	case2.pdf
2026-10-06 15:08 UTC	Risk score computed	System	tier high, score 90
2026-10-06 15:08 UTC	Case status changed	System	submitted → pending_manual_review; automated checks complete

8. Appendix

The detailed technical record, for someone auditing the analysis itself rather than a first-pass reviewer.

8.1 Extracted fields

Normalized values as stored at extraction time. “Located” is the page the value was matched to (used for the purple highlights); a blank means no position was recorded. Arabic values are shown right-to-left, numbers left-to-right.

Document 1 — case2.pdf

Field	Value	Confidence	Located
Issuer	THE GULF INTERNATIONAL PRIVATE أكاديمية الخليج الخاصة ACADEMYO	99%	p.1
Reference number	52086	98%	p.1
Date	2023-07-05	99%	p.1
Amount	57,182.50	98%	p.1
Subtotal	Not found	uncertain	
Tax rate	5.00	95%	p.1
Tax amount	42.50	95%	p.1
Trn	100216382000003	99%	p.1
Student name	GHAYA AHMED SALEM ABDULLA ALKINDI	99%	p.1
Grade	KG2	99%	p.1
Bank beneficiary name	The Gulf International Private Academy	98%	p.1
Bank name	Abu Dhabi Commercial Bank (ADCB)	98%	p.1
Bank account number	260337020001	98%	p.1
Bank iban number	AE250030000260337020001	98%	p.1
Bank branch name	Khaleefa Street, Al Ain, Abu Dhabi	98%	p.1
Parent name	Mr. Ahmed Salem Abdulla AlKindi	98%	p.1
Family number	1245	95%	p.1

8.2 Document hashes (chain of custody)

SHA-256 of each original as recorded when it was uploaded to immutable storage, re-checked against the stored file when this report was generated.

Document	SHA-256	Verification
case2.pdf 59,415 bytes	e9c3ea682c7407ba50d8564f90fb668ee106280583b785e4f80ea151456a70b6	VERIFIED

8.3 Technical parameters used by each check

Page rendering (input to ELA, copy-move, duplicate, visual review)

Render resolution	200 DPI
Colour model	RGB, converted to BGR for OpenCV

Metadata forensics

Libraries	pikepdf / PyMuPDF (PDF structure, XMP, incremental updates)
Modified-after-created rule	flagged if ModDate is more than 1 second(s) after CreationDate
Editing-software names matched	photoshop, gimp, illustrator, affinity, coreldraw, paint.net, pixlr, canva, inkscape

Error level analysis (ELA)

JPEG recompression quality	75
Error gain (scale)	50 (sqrt error term; the reference tool's default)
Contrast stretch / LUT	not applied — the raw error magnitude is thresholded directly
Region threshold	page-relative percentile 99.7 with an absolute floor of 60/255
Region area accepted	0.2% – 60% of the page
Applies to	pages with an embedded raster image only (born-digital text pages are exempt)

Copy-move detection

Keypoint algorithm	BRISK (OpenCV), detector threshold 200
Response / matching / distance thresholds	90 / 20 / 15
Minimum cluster size	30 matching keypoint pairs
Minimum region area	2% of the page
Keypoint / match caps	4,000 / 4,000

Duplicate detection

Hash	perceptual hash (imagehash pHash), 64-bit, one per rendered page
Hamming-distance threshold	5 ($\leq$ this is flagged)
Compared against	every page hash of every previously submitted document

Visual inconsistency / AI-generation review

Model	gpt-4.1-mini
Independent runs per page	2
A finding is kept if	both runs report it, or one reports it with high confidence
Image sent to model	page render, long edge $\leq$ 1536px

Signature / stamp detection and comparison

Model	gpt-4.1-mini
Pages analysed per document	first 10
Output	presence and placement only; comparison is a qualitative verdict, never a numeric score

Field validation

Sub-checks	date not in the future; subtotal vs line items; total vs subtotal + tax; reference-number format
Amount tolerance	1% of the amount, minimum 0.01

Issuer verification

Matching	rapidfuzz weighted ratio against the active issuer registry, with a model-judged fallback for cross-script names
Similarity threshold	85

Field locations (purple highlights)

Method	extracted values matched back onto OCR word positions (Azure Document Intelligence Layout)
Repeated values	resolved by nearby keywords (“total”, “subtotal”, “VAT”, “date”...)
Accuracy	approximate; a field that cannot be matched is reported without a highlight

9. PDF Highlighted Regions

Every flagged page, rendered from the original with its findings drawn in. These are copies made for this report; the original files are untouched. Only pages with at least one highlight appear. Each highlight has a region ID that Section 5 refers back to.

Colour key

	ELA (solid red)
	Copy-move (solid orange)
	Font mismatch in the text layer (solid fuchsia)
	Field exception (solid purple)
	Model judgment (dashed blue, approximate)
	Page-level finding (not localized)

Region index

Region	Document	Page	What it shows
R1	Document 1 (case2.pdf)	1	Visual review — text alignment (approximate)
R2	Document 1 (case2.pdf)	1	Duplicate detection — near-duplicate page
R3	Document 1 (case2.pdf)	1	Field exception: Tax amount (42.50 vs 5% = 2,857.00)
R4	Document 1 (case2.pdf)	1	Field exception: Tax rate (5% gives 2,857.00)

case2.pdf — page 1

Section 9 · Document 1 of 1 · regions R1, R2, R3, R4

Field exception (solid purple) Model judgment (dashed blue, approximate) Page-level finding (not localized)

TAX INVOICE

Inv. : 52086
TRN : 100216382000003
Date : 05/07/2023

The Gulf International Private Academy in Al Ain certifies that the below student(s) is/are registered in our school for the academic year 2023-2024. Fees for registered students(s) are as follows:

Name	Grade	Tuition Fees	Books	Uniform	Technology	Other	5%	Total
GHAYA AHMED SALEM ABDULLA ALKINDI	KG2	52,130	000	200	550	3,510	42.50	57,182.50

Bank Detail:

Beneficiary Name:	The Gulf International Private Academy
Beneficiary Bank Name:	Abu Dhabi Commercial Bank (ADCB)
Beneficiary Account Number:	260337020001
Beneficiary IBAN Number:	AE250030000260337020001
Beneficiary Branch Name:	Khaleefa Street, Al Ain, Abu Dhabi
New Email ID:	Accountant@gina.ae

This letter has been issued upon the guardian's request without any liabilities for our school. The fees stated in this document are in accordance to the Abu Dhabi Department of Education and Knowledge (ADEK) approved school fees.

Parent Name : Mr. Ahmed Salem Abdulla AlKindi
Family Number : 1245

Registration Department **Accounts Department**

R1 · Visual review — text alignment (approximate)

Potential Growth Here

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R1. Visual inconsistency / AI-generation review

Page 1 — Text alignment: The signatures at the bottom are misaligned and have different baselines compared to the typed text around them, notably in the Registration Department and Accounts Department areas (approx. bounding box x:0.1, y:0.8, width:0.4, height:0.15). This is a vision-model judgment, not pixel-level analysis — a probabilistic signal, not a definitive finding.

R2. Duplicate detection (page-level — not localized to a region)

Page 1 is a near-identical match (perceptual hash Hamming distance 0) to page 1 of "sample21.pdf" in case CASE-CF609781 — a possible duplicate or near-duplicate resubmission.

R3. Field validation

Tax 42.50 is not 5% of the total less tax 57,140.00 (expected 2,857.00). (Tax amount: 42.50)

R4. Field validation

Tax 42.50 is not 5% of the total less tax 57,140.00 (expected 2,857.00). (Tax rate: 5%)